File Handling

file.readlines()

→ reads all lines of a file into an identifier as an iterable.

① Read a whole file

```
with open('eg.txt', 'r') as file:
    content = file.read()
    print(content)
```

→ reads the entire file.

⇒ prints all the content in the eg file.

② Read a file line by line

```
with open('eg.txt', 'r') as file:
    for line in file:
        print(line)
```

→ iterating through the file

⇒ prints all the lines, including the newline character.

So → `print(line.strip())`

③ Write into the file.

write mode

```
with open('eg.txt', 'w') as file:
    file.write('Hello World! \n')
    file.write('This is a new line! \n')
```

⇒ writes 2 lines into the eg.txt.

\n → newline character.

Note: This method overwrites everything in the eg. file.

④ Appending data into a file

```
with open ('eg.txt', 'a') as file:
    file.write ('This is new data')
```

append mode

⇒ writes/appends the new data into the eg.txt, after the old data.

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⑤ Writing a list of lines into a file.

```
lines = ['a...\n', 'b...\n', 'c...\n']
with open ('eg.txt', 'a') as file:
    file.writelines (lines)
```

→ writelines ( )

⇒ g.txt

```
a...
b...
c....
```

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⑥ Binary files (.bin files)

Reading	Writing

`with open ('eg.bin', 'rb') as f:`
`c = file.read()`
`print (c)`
	`with open ('eg.bin', 'wb') as f:`
	`file.write (data)`

read byte mode	write byte mode
⇒ writes data into binary file	

usually hexadecimal

w+ mode & cursor move (seek())

```
with open('eg.txt', 'w+') as file:
    file.write("Hello world!\n")
    file.seek(0)
    file.read()
```

→ write + read mode

→ writes the content into the eg.txt file

→ moves the pointer to the beginning of the file

→ reads everything that is available ahead of the pointer.

CWD = Current Working Directory

(A) Managing file paths

(1) Creating a new directory.

```
import os

new_dir = 'packages'

os.mkdir (new_dir)
```

⇒ creates a new folder 'packages' in the CWD

(2) Listing all items in the CWD

```
items = os.listdir ('.')
print (items)
```

→ for cwd.

(3) Joining paths

```
dl = 'folder'
file = 'eg.txt'

cwd = os.getcwd()

full_path = os.path.join(cwd, dl, file)
```

dl — folder name
file — file name
cwd — current working directory path.

in order

PTO

(4) Checking if a path exists.

```
path = 'eg.txt'
if os.path.exists(path):
    print('Yes')
else:
    print('No')
```

condition with os.path.exists()

(5) Checking if a path is a file / dir / neither.

path = 'eg.txt'

os.path.isfile(path) → returns True if path is a file in CWD.

os.path.isdir(path) → returns True if path is a folder in CWD

(6) Getting the absolute path

relative_path = 'eg.txt'

absolute_path = os.path.abspath(relative_path)

or full path

→ returns the absolute path of the given path.

os.path.abspath()

(7) Deleting a directory (os.rmdir())